



September 3rd, 2026

Confirmation of participation and support for the publication of a book on *high-speed electrical machines*

Dear Editors,

The plan for this book, and the part I would take into it, has been described to me. I am writing to support the proposal and to confirm that I intend to contribute to the book.

My own work is on the losses in electrical machines and on the magnetic materials that produce them; from that position, the gap the authors describe is one I meet constantly. The loss models that machine designers rely on are fitted to measurements made at or near mains frequency, and they are then applied, usually without much hesitation, at fundamental frequencies ten or twenty times higher; they do not extrapolate. The separation into hysteresis, eddy-current and excess components loses much of its meaning as the frequency rises, and the degradation that cutting, punching and stacking inflict on the material is itself frequency dependent, so the correction factors in common use have no dependable basis in the high-speed range. An engineer who wants to do better than a safety factor must assemble the method from scattered journal papers; there is no book that sets it out.

As I understand it, that is what this book addresses, and it does so as a design book rather than as a survey; I consider that the right decision. The review literature on high-speed machines is already substantial, and a further review would just add little. What the field lacks is a text that tells an engineer what to do when the ordinary rules stop holding and that treats the mechanical, thermal, and converter constraints as part of the same problem rather than separate specialisms.

Subject to the publishing contract, I intend to contribute the loss chapters of the book, on AC winding losses and conductor design and on the core, rotor and aerodynamic losses that arise at high speed, together with members of my research group at Aalto University. I have worked on loss measurement and modelling, the characterisation of magnetic materials, and the finite-element analysis of electrical machines for over 25 years and would bring the results of our recent research projects to those chapters.

The book is proposed as a companion to “*Design of Rotating Electrical Machines*”, which is a standard reference in our field. A volume that carries the same reader into the regime where that book's rules cease to apply would be genuinely useful, and I am glad to take part in it.

Sincerely,

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